

YOHAN SEQUEIRA

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EDUCATION

Virginia Tech — Ph.D., Engineering Science and Mechanics

Expected Aug 2028

Cornell University, CALS — M.S., Biological and Environmental Engineering

Aug 2020

Cornell University, CALS — B.S., Biological and Environmental Engineering

May 2019

RESEARCH EXPERIENCE — SENSING & INSTRUMENTATION, BIOMECHANICS

Open-Ocean Stereo-Camera Tracking System, Flying Squid | *Researcher*

Tokai University, Japan | 2024–2025

- Designed, built, and ruggedized a custom multi-camera stereo sensing system (camera-and-mirror custom optical configuration) for open-ocean field deployment, capturing 3D flight trajectories of free-flying squid from a research vessel.
- Developed a novel mathematical framework and MATLAB toolbox for calculating stereo measurement error as a function of range and camera parameters, generalizable across stereo-vision system designs for multi-camera systems.
- Built a multi-sensor fusion and registration pipeline (IMU accelerometer/gyroscope, point-cloud, magnetometer, GPS) to filter data across independent, asynchronous sensor streams for the purpose of aligning reconstructed flight trajectories.
- Calibrated a wind tunnel and collected aerodynamic force data on reference airfoils and flying squid models in various configurations to validate aerodynamic predictions and provide data for static and dynamic stability models of squid flight.

Flight Dynamics & Control, Entada Seed Autorotation | *Researcher*

Virginia Tech, Blacksburg, VA | 2024–2026

- Characterized dynamic flight modes of autorotating seed structures using high-speed videography and 3D motion tracking.
- Designed and 3D-printed robotic seed models with tunable mass distribution to physically validate dynamical predictions.
- Built coupled aerodynamic and rigid-body dynamics models linking morphology and mass distribution to flight-mode stability and control authority and identify transitions between stable flight modes (autorotation, tumbling, fluttering).

Self-Righting Dynamics & Robotics, Benthic Crabs | *Researcher*

Friday Harbor Labs, WA | 2024–2026

- Set up, calibrated, and iterated on underwater stereo-camera systems to capture righting kinematics of crabs.
- Designed and built self-righting robots capable of multi-environment righting behavior (in air and on a free surface).
- Developed mathematical models of self-righting dynamics to characterize passive vs. active stabilization strategies across body morphologies. Extended mathematical models to cover various self-righting strategies across animal species.

Deep-Sea Multi-Camera Sensor System (MINION) | *Intern and Researcher*

MBARI, Moss Landing, CA | 2021–2022

- Designed a synchronized multi-camera stereo sensor system rated to 500 m depth for particle-field imaging.
- Reduced system cost by designing 3D-printed pressure housings and using a Raspberry Pi-based control system.
- Validated optical design using Blender-based light-transport simulation to confirm imaging performance across targets.

High-Speed Camera & Acoustic Array, Bat Flight | *Researcher*

University of Brunei Darussalam | 2021–2023

- Designed a 50-channel high-speed camera array and a 32-channel ultrasonic acoustic array to capture synchronized kinematic and acoustic data of live bats flying and maneuvering in laboratory conditions.
- Developed semi-automated analysis pipelines (visual-hull ray tracing, automated camera calibration, deep-learning-based audio analysis) to extract quantitative data from raw sensor output, integrated pipelines into a simplified application.

ADDITIONAL RESEARCH AND PROJECT EXPERIENCE

Perching Drone Design and Optimization | *Researcher*

Virginia Tech, Blacksburg, VA | 2026

- Designed a 5-inch quadcopter frame MAV carrying durable filament, able to unspool, wrap and hook onto filament lines
- Computed capstan-friction values for various material combinations (filament and post-substrate) to optimize wrap strength

Bio-Inspired Flight Vehicle Design (Chiropter) | *Team Mentor*

Virginia Tech, Blacksburg, VA | 2021–2023

- Advised on design, prototyping, and manufacture of a flapping-wing bio-robotic flight vehicle.
- Evaluated and integrated alternative power-electronics and wing-membrane solutions to improve vehicle robustness.

PUBLICATIONS, PRESENTATIONS & GRANTS

[1] Sequeira, Y. Machine learning methods for reconstructing the acoustic fields of bat biosonar. ASA, Dec 2022.

[2] Sequeira, Y. et al. Characterizing the Marangoni effect in NASA STDCE-1: Inspiration to self-propelling robots. APS DFD, Mar 2021.

[3] Sequeira, Y. et al. Revisiting the NASA Surface Tension Driven Convection Experiments. NPJ Microgravity, Feb 2020.

[4] Sequeira, Y., Socha, J. When Squid Fly: Encapsulating the anatomy of the flying squid in a working robotic model. SICB, 2024.

[5] Sequeira, Y., Carrington, E., Socha, J. Self-righting behavior of benthic crabs in freefall. SICB, 2025.

[6] Sequeira, Y. et al. Falling with a spin: the dynamic flight of Entada seed capsules. SICB, 2026.

Grants and Awards: Author, Flying Squid Field Research Grant, \$10,000 from VT MATHBIO Research Program. Recipient, Crab Self-Righting Research Study Award, Friday Harbor Labs Student Research Fellowship, \$3,000. Recipient, BIOTRANS rotation program for graduate students, one-year stipend and ongoing travel funding.

TECHNICAL SKILLS

Sensing & Instrumentation: Stereo-vision & multi-camera systems (OpenCV, DLT-based methods, custom stereo rigs), high-speed videography (Photron, Edgetronic, Phantom), CT/3D scanning (micro-CT, structured-light scanning), microphone array design (ultrasonic transducers, National instruments DAQ), stereophotogrammetry (DLTdv8, MATLAB)

Design & Fabrication: Rapid prototyping (FDM/SLA 3D printing, laser cutting), CAD (Fusion 360, Siemens NX, CATIA), Arduino and ESP32 microcontrollers (I2C/SPI sensor integration), 3D modeling (Blender, Meshlab)

Modeling & Analysis: Flight dynamics modeling (rigid-body, aerodynamic coupling), MATLAB (image processing, dynamics, signal processing, deep learning, reinforcement learning), Python (TensorFlow, PyTorch, SciPy, scikit-learn, pandas), PIV/CFD analysis (PivLab, COMSOL)